Thin Film Logbook

One place for every sample, step, analysis and data file

October 22, 2026

CTN

Why a logbook?

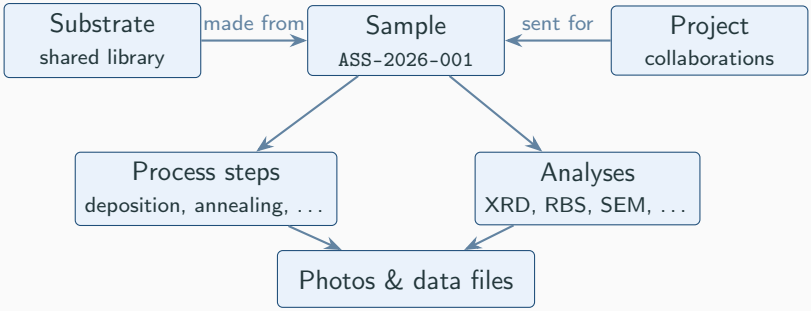
Today

- Paper notebooks and personal spreadsheets
- Data files scattered across instrument PCs and USB sticks
- "Which substrate was that? What was the RTA temperature?"
- Hard to hand samples over or find a colleague's results

With the logbook

- A unique ID and a full history for each sample
- Parameters entered in structured fields, so they're searchable
- Photos and raw data attached where they belong
- Shared, searchable, exportable to CSV

How it's organised



Steps and analyses share one timeline per sample.

Sample IDs

ASS-2026-001-A-2

initials year counter piece piece of piece

`ASS-2026-001`	ASS's first sample of 2026
`ASS-2026-001-REF`	a reference piece of it
`UMIN1-2026-001`	a collaboration sample (project code)

Generated automatically and never changed. Use the **Label** for your own description.

Four kinds of sample

Thin film	a library substrate + dimensions; film material once deposited
Bulk	a piece of a library substrate; external ID if someone else cut it
Commercial	description, supplier, batch number, spec sheet required
Collaboration	sent by a partner institution: its project, a description, their ID

All of them: optional label and tags, and a researcher (you, or someone you create it for).

Recording a step or analysis

- Pick the type, and the form shows its fields
- **Where** + **Equipment**: only the machines at that place
- Units built in; `1e-4` and `12:30` both work
- Dropdowns remember values; + **Add new…** for a new one
- **Performed by**: whoever ran it can see & edit that entry

Steps: Deposition, Annealing (Furnace / RTA), Ion Implantation, Lithography, Etching, Cut, Other

Analyses: XRD / HRXRD (with modes), XPS, SEM, TEM, RBS, AFM, Raman, Optical Microscopy, Hall / 4-probe, Ellipsometry, Profilometry, Other

Admins can add types and fields without touching code.

Photos and data files

Add photos

- Opens the phone camera at the bench
- Several at once, or drag onto a card
- Resized to ≤ 2 MB, HEIC converted, GPS removed

Attach files

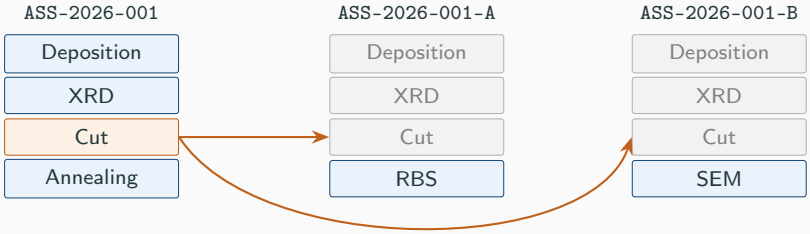
- Raw data, stored exactly as uploaded
- 16-bit TIFFs, `.raw`, `.csv`, PDFs, …
- Thumbnails on the timeline; one folder per sample on disk

One run, several samples

Same deposition run or same XRD session for several samples?

- Fill in the form once, tick **Also add to other samples**
- Each sample gets its own linked copy (“Also on: . . .”)
- Shared photos are stored once, and each sample can get its own data file
- Edits can be applied to the whole run, with a warning before anything another sample has would be overwritten

Cutting a sample into pieces



- Pieces inherit the history up to the cut (grey), so a correction on the parent reaches every piece
- Each piece then has its own entries; pieces can go to other people
- Optional reference piece `-REF`

Who sees what

Everyone: each sample's summary

- ID, label, type, materials, substrate
- tags, techniques used, researcher

Details (parameters, notes, files) are visible to:

- the owner and admins
- people it's shared with: **View** or **Edit**
- for one entry, whoever performed it

Samples can be handed over to another researcher; they confirm with **Accept**/**Decline**, and the ID stays the same.

Search and export

- Filter by text, researcher, type, material, technique, tag
- Group by researcher, year, type, material, technique, tag, project
- **Export results (CSV)**: one row per step/analysis, one column per parameter, so it loads straight into Origin, Excel or pandas
- Select several samples and share them in one go

Also

- **Substrate library**: specs and spec sheets entered once, reused
- **Projects**: partner institutions and their samples
- **Suggestions**: report a bug or ask for a field, and admins reply
- Portuguese / English, dark mode, works on phones

Getting started

1. Ask an admin for an account
2. Sign in, set your password, name and initials
3. Create a sample: **New Sample**
4. Add steps and analyses as you go, with photos from your phone

The user manual (`docs/manual`) has the details.

Questions?